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List of Abbreviations

CCP Chance-Constrained Program

CVaR Conditional Value at Risk

DRO Distributionally Robust Optimization

ECP Expected Constraint Program

EIC European Innovation Council

MAS Multi-agent System

STL Signal Temporal Logic

WP Work Package

1. Introduction

This report is the deliverable for Task 2.4 in the second year of the SymAware project. It is based on the following research contributions:

- A. Bahari Kordabad, E. E. Vlahakis, L. Lindemann, D. V. Dimarogonas, and S. Soudjani, "Distributionally Robust Control for Chance-Constrained Signal Temporal Logic Specifications," *IEEE Conference on Decision and Control (CDC)*, 2024 [accepted].
- E. E. Vlahakis, L. Lindemann, P. Sopasakis, D. V. Dimarogonas, "Probabilistic Tube-based Control Synthesis of Stochastic Multi-Agent Systems under Signal Temporal Logic," *IEEE Conference on Decision and Control (CDC)*, 2024 [accepted].
- "Multi-agent Distributionally Robust Risk-based Control for Signal Temporal Logic," *preparing for a journal publication*, [under progress].

These works focus on advanced techniques in distributionally robust control and risk-based control frameworks, with a particular emphasis on addressing Signal Temporal Logic (STL) specifications under conditions of uncertainty and limited information. Conducted as collaborative efforts between MPI-SWS and KTH, with MPI-SWS leading the research, these studies contribute to the objectives of the SymAware project. They advance both the theoretical foundations and practical methods for task planning and control in complex, multi-agent environments where available information is restricted.

1.1 SymAware: High-level Description of the Project

SymAware addresses the fundamental need for a new conceptual framework for awareness in multi-agent systems (MASs) that is compatible with the internal models and specifications of robotic agents and that enables safe simultaneous operation of collaborating autonomous agents and humans. The goal of SymAware is to provide a comprehensive framework for situational awareness to support sustainable autonomy via agents that actively perceive risks and collaborate with other robots and humans to improve their awareness and understanding while fulfilling complex and dynamically changing tasks. The SymAware framework will use compositional logic, symbolic computations, formal reasoning, and uncertainty quantification to characterise and support situational awareness of MAS in its various dimensions, sustaining awareness by learning in social contexts, quantifying risks based on limited knowledge, and formulating risk-aware negotiation of task distributions. These objectives will be achieved in SymAware through

1. Logical characterization of awareness using symbolic methods,

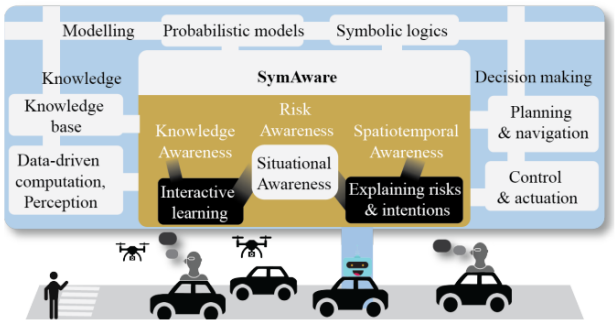

Figure 1.1: The novel cognitive architecture developed in SymAware.

2. Quantifying the symbolic reasoning for awareness with spatial and temporal ingredients for decision making,
3. Risk awareness via quantified knowledge,
4. Quantifying and communicating knowledge awareness,
5. Demonstrating awareness engineering in aviation and automotive use cases, and
6. Identifying requirements for ethical and trustworthy awareness in human-agent interaction.

The objectives of SymAware address the "Awareness Inside" Challenge of the European Innovation Council (EIC) by extending and formalizing human-based models of situational awareness and by providing a novel conceptual situational awareness framework for MASs that encompasses logical characterisation and integrative formal reasoning of interdependent awareness dimensions including knowledge, spatiotemporal, risk and social dimensions. This will support transitioning to safe mixed operation of autonomous agents and humans.

Figure 1.1 shows a schematic view of the SymAware new cognitive awareness architecture. Together the objectives of this proposal will achieve the depicted situational awareness for multi-agent settings in which the situational awareness is related to the different awareness dimensions and to contemporary methods for probabilistic modelling, perception, estimation, and (data-driven) computation.

Figure 1.2 shows the main technical work packages of SymAware. The comprehensive framework of situational awareness and its logical characterisation (Objective 1) will be developed in Work Package (WP) 1. The spatiotemporal awareness of Objective 2 will be studied in WP2. Risk and knowledge awareness (Objectives 3 and 4) will be studied in WP3 and WP4, respectively. The automotive and aviation use cases (Objective 5) will be developed in WP5. Finally, the human-robot interaction and the ethical requirements (Objective 6) will be studied in WP6.

1.2 Purpose and Contents of This Report

This report is the fourth deliverable of WP2 on spatiotemporal awareness and specifically on Task 2.4 about task planning under limited information. The report is organized as follows: Chapter 2 explores spatiotemporal awareness and the use of STL for managing time-sensitive tasks,

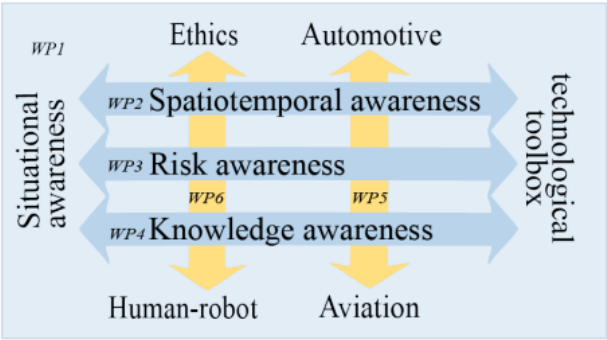

Figure 1.2: The main (technical) work packages of SymAware.

while addressing communication and sensing limitations in multi-agent systems. Chapter 3 covers decentralized control synthesis, stochastic control synthesis of multi-agent systems under STL, and Distributionally Robust Optimization (DRO) for decision-making under uncertainty, including DRO applications to STL specifications and Chapter 4 delivers conclusions.

2. Task Planning with Spatiotemporal Awareness

In this section, we describe an overview of spatiotemporal awareness in task planning for dynamical systems in interaction with environment and/or other agents. Spatiotemporal awareness in task planning for dynamical systems, especially in multi-agent settings, is essential for ensuring that agents can successfully complete tasks that are subject to both spatial and temporal constraints. In such systems, agents interact by exchanging information, often constrained by communication and sensing limitations. These limitations can affect the execution of high-level tasks defined by spatiotemporal logic, such as Signal Temporal Logic (STL), which provides a precise framework for specifying time-sensitive and spatial rules. When mismatches arise between task dependencies and the sensing/communication network, spatiotemporal awareness metrics quantify how well agents satisfy their tasks. These metrics can inform adaptive motion planning strategies that reconfigure agent behavior in real time, either by finding least-violating solutions or by adjusting task specifications to maximize robustness, thus enhancing the system's overall performance despite dynamic challenges.

2.1 Task Specification: Signal Temporal Logic

Control synthesis for dynamical systems against complex tasks has recently garnered significant attention due to the increasing demand for autonomous systems that can handle sophisticated (potentially time-varying) objectives and constraints [1]. One prominent approach to addressing this challenge involves leveraging temporal logic, a well-established method for encoding complex specifications, particularly those involving spatial and/or temporal requirements for dynamical systems. Temporal logic serves as a powerful framework for ensuring that the desired behaviors of a system are realized within specified time deadlines while respecting spatial constraints, making it suitable for a wide range of applications from robotics to autonomous vehicles [2]. The use of temporal logic in control synthesis has therefore emerged as a promising avenue for developing control policies that guarantee complex, high-level system behaviors.

STL, in particular, stands out as a formal language that allows for the encoding of time-constrained tasks by providing a closer connection to the physical system. STL achieves this by offering robust semantics that map directly to system trajectories, meaning that the satisfaction of the task can be evaluated in a quantitative manner [3]. This robustness provides a means to not only determine whether a specification has been met but also how well it has been met, facilitating the design of systems that are robust to uncertainties.

To understand STL more intuitively, consider it as a way to describe how a system should behave over time. Instead of simply specifying static conditions for a system to meet, STL allows us to define when certain conditions should be met and for how long they need to hold. For example, we might want a robot to reach a specific point within a certain time window or avoid a dangerous area during its operation. These types of time-sensitive and spatially aware tasks are naturally expressed using STL.

More specifically, STL is a predicate logic composed of atomic predicates μ that may be false $\perp$ or true $\top$ [4]. The assessment of the validity of each atomic predicate μ relies on the sign of a continuously differentiable predicate function $h : \mathbb{R}^n \to \mathbb{R}$ as follows:

$$\mu = \begin{cases} \top, & \text{if } h(x) \geq 0 \\ \perp, & \text{if } h(x) < 0\,, \end{cases}$$

where $x \in \mathbb{R}^n$ is the state of the system.

Semantics: The syntax of the STL formulae is defined recursively according to the following grammar:

$$\phi ::= \top \,|\, \mu \,|\, \neg\phi \,|\, \phi \wedge \psi \,|\, \phi\,\mathcal{U}_{[a,b]}\,\psi\,,$$

where ϕ, ψ are STL formulas, notations $\neg$ and $\wedge$ denote *negation* and *conjunction* of formulas and $\mathcal{U}_{[a,b]}$ denotes the *until* operator. The interval $[a, b]$ represents the time period during which the specification must be satisfied.

Negation ($\neg$) simply means that if a statement is true, then its negation is false, and vice versa. For instance, if a system is in a safe region, the negation would state that the system is *not* in the safe region. Conjunction ($\wedge$), on the other hand, refers to the logical AND operation: both statements must be true for the entire expression to be true. For example, we might want the system to avoid an obstacle (predicate μ_U) *and* reach a target region (predicate μ_T) within a certain time frame. In such a case, both conditions must hold simultaneously.

The *eventually* and the *always* operators are defined as $\mathcal{F}_{[a,b]}\phi := \top\,\mathcal{U}_{[a,b]}\,\phi$ and $\mathcal{G}_{[a,b]}\phi := \neg\mathcal{F}_{[a,b]}\neg\phi$, respectively. The satisfaction relation $(\xi, k) \models \phi$ denotes if the sequence ξ satisfies ϕ at time k, where $(\xi, k) := \{x_k, x_{k+1}, \ldots\}$ is a trajectory of a discrete-time system starting at time k. Note that these definitions can also be developed to the continuous-time dynamics.

Semantics: The satisfaction of an STL formula ϕ by a trajectory ξ at time k is defined recursively as follows:

$$\begin{aligned}
&(\xi, k) \models \mu && \Leftrightarrow \quad h(x_k) \geq 0 \\
&(\xi, k) \models \neg\phi && \Leftrightarrow \quad \neg((\xi, k) \models \phi) \\
&(\xi, k) \models \phi \wedge \psi && \Leftrightarrow \quad (\xi, k) \models \phi \wedge (\xi, k) \models \psi \\
&(\xi, k) \models \phi\,\mathcal{U}_{[a,b]}\,\psi && \Leftrightarrow \exists k' \in \{k+a, \ldots, k+b\}, (\xi, k') \models \psi \wedge \forall k'' \in \{k, \ldots, k'\}, (\xi, k'') \models \phi\,.
\end{aligned}$$

Example: A simple reach-avoid specification can be formulated as follows:

- Reach the target set T within the time interval $[a, b]$: $\mathcal{F}_{[a,b]}\mu_T$, where μ_T is the predicate that is true when the system is within the target set T.
- Avoid the unsafe set U over the entire interval $[0, b]$: $\mathcal{G}_{[0,b]}\neg\mu_U$, where μ_U is the predicate that is true when the system is within the unsafe set U.

Thus, the complete reach-avoid specification is:

$$\phi_{\text{reach-avoid}} = \mathcal{G}_{[0,b]}\neg\mu_U \wedge \mathcal{F}_{[a,b]}\mu_T$$

This formula expresses that the system must avoid the unsafe set U at all times during the interval $[0, b]$, while eventually reaching the target set T within the time interval $[a, b]$. See Figure 2.1.

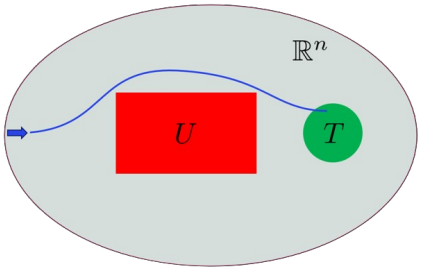

Figure 2.1: A reach-avoid task.

This formal language provides a structured and rigorous way to model spatiotemporal awareness in the task planning of multi-agent systems, which is critical in ensuring that these systems can collaborate effectively while adhering to complex specifications. The importance of STL in this context lies in its ability to encode and verify the temporal and spatial interactions between agents, thereby supporting the design of systems that operate in dynamic and unpredictable environments. By offering a systematic approach to handling time-sensitive and spatially dependent tasks, STL has become an essential tool in advancing the capabilities of multi-agent systems.

2.2 Multi-Agent Task Planning and Information Exchange

In multi-agent systems, an informed task planning and coordination requires effective communication and information exchange between agents. However, real-world limitations often arise due to constraints in the ability to exchange information. These constraints could be due to limited communication bandwidth, network congestion, or delays in transmission. Additionally, agents might have incomplete information because of sensing limitations, where not all agents are equipped with advanced sensors, or the environment may obstruct their sensing capabilities. These factors can lead to situations where agents must make decisions based on outdated or incomplete data, impacting their ability to collaborate effectively. Therefore, multi-agent task planning often requires strategies to manage and work around these information gaps.

2.2.1 Communication and Sensing Constraints

One major challenge in multi-agent systems is that communication and sensing constraints can significantly affect the performance of the team [5, 6] (see Figure 2.2), for example, when the specifications among agents are modeled by STL [7]. Due to bandwidth limitations, agents might only be able to share a limited amount of data at any given time, forcing them to prioritize what information is most critical. For instance, agents may need to limit the frequency of updates or focus on sharing only high-priority information, such as potential hazards or significant environmental changes. Furthermore, there can be delays in the exchange of information due to network latency or interference, which can result in agents acting on outdated information. These communication bottlenecks can prevent agents from having a complete and up-to-date understanding of their shared environment, which is crucial for effective collaboration.

Figure 2.2: An informed multi-agent task planning needs an effective information exchange.

2.2.2 Decision Making with Limited Information

To overcome these communication and sensing challenges, multi-agent systems often rely on coordination strategies designed to minimize the impact of limited information. One such strategy is decentralized planning, where agents make decisions locally based on the information available to them, rather than relying on constant updates from other agents. Another approach is to employ robust task-planning algorithms that account for uncertainty, allowing agents to proceed with tasks even when some information is missing. Additionally, agents can adopt consensus-based methods, where they periodically share critical information with their neighbors to ensure that the entire group is on the same page. These strategies help the agents coordinate effectively despite the inherent limitations in communication and sensing.

It is unrealistic to expect that agents will have access to infinite or perfect information due to various practical constraints. Factors such as energy limitations, processing power, and communication infrastructure all impose restrictions on how much information can be shared and processed at any given time. In addition, environmental factors such as obstacles, weather conditions, and interference can further limit the ability of agents to communicate and sense their surroundings. As a result, multi-agent systems must be designed with these limitations in mind, balancing the need for information with the practical realities of real-world operations. Successful multi-agent coordination hinges on developing efficient communication protocols and planning algorithms that optimize the use of available information while ensuring that agents can still achieve their collective goals.

2.3 Conclusion

In conclusion, spatiotemporal awareness in task planning for dynamical systems, particularly in multi-agent settings, proved to be essential for achieving effective coordination and task execution under spatial and temporal constraints. This chapter introduced Signal Temporal Logic (STL) as a robust framework for defining and managing time-sensitive and spatially dependent tasks. STL provided a systematic approach to formulate these complex tasks. The challenges associated with limited information necessitated strategic approaches to decision-making, such as robust planning,

which enabled agents to adapt and coordinate effectively even with incomplete data. By acknowledging the practical constraints on information exchange and processing, the chapter underscored the need for efficient protocols and adaptive algorithms.

For instance, in applications like autonomous vehicle coordination, MAS can successfully meet objectives such as collision avoidance and route optimization by leveraging spatiotemporal logic to ensure each vehicle navigates safely while meeting time-bound goals. Similarly, MAS in drone fleets can handle tasks like area coverage or search-and-rescue operations by coordinating movement and task allocation, adhering to spatial and temporal requirements, even when communication or sensing data is limited.

These examples are directly linked to the broader themes explored in this report. The report first addressed the importance of STL in encoding complex specifications under limited information, followed by a discussion of decentralized planning techniques and robust task-planning algorithms, which were shown to be critical for MAS facing communication and sensing constraints. This connection highlights how agents can still contribute to global objectives while making local decisions based on incomplete data.

In addition, the report examined control synthesis under stochastic noise, using a tube-based constraint-tightening approach that allows relaxation of stochastic control problems into deterministic ones with tighter constraints. The multi-agent system example demonstrated how this method could scale to larger systems. Moreover, optimization approaches like Distributionally Robust Optimization (DRO) were presented to address uncertainty by providing probabilistic guarantees, even with incomplete data, ensuring robust decision-making in dynamic environments.

These considerations ensured that multi-agent systems could meet their objectives and maintain high performance in dynamic environments.

3. Task Planning using Limited Information

Task planning under limited information demands a robust control synthesis to effectively manage and adapt to unseen or incomplete data. In situations where full information about the environment or task requirements is unavailable–due to factors like restricted communication channels, limited sensors, or dynamic conditions–control systems must be designed to handle uncertainty and adapt to emerging data. This requires employing sophisticated algorithms that can operate effectively with partial information and dynamically adjust as new data becomes accessible. Techniques such as decentralized decision-making, where agents make informed choices based on available local information and collaborate with others, alongside robust planning methods that anticipate potential data gaps, are essential. These strategies ensure that systems can maintain performance and achieve their goals even when confronted with incomplete or evolving information. In this chapter, we briefly explain decentralized control synthesis and Distributionally Robust Optimization (DRO) as two key techniques for decision-making under limited information.

3.1 Decentralized Control Synthesis

Decentralized control synthesis is a technique used in large scale interconnected systems where control decisions are made independently by each agent based on local information rather than relying on a central authority [8]. This approach distributes the decision-making process among multiple agents, allowing them to operate more autonomously while still working towards a common objective. The decentralized control synthesis contrasts with the centralized control, where a single central authority or organization makes all major decisions and manages operations. By decentralizing control, systems can achieve greater scalability and flexibility, as each agent can adapt to changes in the environment and the behavior of other agents in real time. This technique is particularly useful in complex large scale systems where a central controller might be impractical.

In the context of multi-agent systems using Signal Temporal Logic (STL), decentralized control synthesis becomes even more relevant [9]. STL provides a framework for specifying complex temporal and spatial requirements, ensuring that agents adhere to certain behaviors and constraints over time. Decentralized control synthesis allows each agent to interpret and implement these STL specifications based on its local perspective and communication with neighboring agents. This approach ensures that the global objectives defined by STL are met even when each agent operates with only partial knowledge of the overall system. By breaking down the control problem into manageable, local tasks, agents can effectively coordinate their actions to satisfy the STL requirements,

ensuring that the system as a whole adheres to the desired spatiotemporal constraints.

Decentralized control synthesis is particularly advantageous for decision-making under limited information. When agents only have access to partial or delayed information, centralized control systems may struggle to maintain an accurate global perspective. In contrast, decentralized control allows each agent to make decisions based on its local observations and interactions, thereby mitigating the impact of missing or outdated information. By focusing on local data and fostering communication among agents, decentralized control helps maintain system performance and adaptability. This approach enables the system to respond dynamically to new information and uncertainties, ensuring that decision-making remains robust even when complete information is not available.

In the following, we consider the control synthesis of stochastic linear Multi-Agent Systems (MASs) under signal temporal logic specifications permitting multiple individual and joint tasks per agent, with the global specification expressed as a chance constraint with a specified probability. We first present a probabilistic tube-based control synthesis to address this problem when agents' dynamics are perturbed by additive stochastic noise the distribution of which is partially known. All proofs of the presented results can be found in [10]. Subsequently, we summarize a distributionally robust optimization approach to solving this problem when the probability distributions of the uncertain parameters lie in ambiguity sets.

3.2 Stochastic Multi-Agent Systems under Signal Temporal Logic

Here, we solve a stochastic optimal control problem of a discrete-time linear MAS subject to additive stochastic perturbations and a global STL specification permitting multiple individual and joint tasks per agent. First, we decompose the multi-agent dynamics into a deterministic system and an error closed-loop stochastic system, for which we construct a probabilistic reachable tube (PRT) [11] as the Cartesian product of reachable sets of individual error systems. These are driven by stochastic disturbances lying in confidence regions (CRs) with a fixed probability. By assuming independence among individual disturbances, we show that the PRT probability can be controlled by the product of probabilities selected for each individual CR and a union-bound argument applied over time. Thus, by lower bounding the PRT probability by the specification probability, we can tighten all state constraints induced by the STL specification by solving tractable optimization problems over segments of the PRT. For multi-agent STL specifications, this is a less conservative alternative to tightening approaches relying on the STL structure [12, 13]. An attainable feasible solution to the resulting deterministic problem can then be used to synthesize multi-agent trajectories that satisfy the STL specification with the desired confidence level. Subsequently, to enhance scalability, we decompose the resulting deterministic problem into agent-level subproblems, which are solved iteratively according to a scheduling policy. We show that this iterative procedure is recursively feasible, ensures satisfaction of local tasks, and guarantees nondecreasing robustness for joint tasks. We present our method in the following paragraphs starting with the notation used in this section.

3.2.1 Notation and Preliminaries for Section 3.2

The sets of real numbers and nonnegative integers are $\mathbb{R}$ and $\mathbb{N}$, respectively. Let $N \in \mathbb{N}$. Then, $\mathbb{N}_{[0,N]} = \{0, 1, \ldots, N\}$. Let $x_1, \ldots, x_n$ be vectors. Then, $x = (x_1, \ldots, x_n) = [x_1^\intercal \ \cdots \ x_n^\intercal]^\intercal$. We denote by $\boldsymbol{x}(a : b) = (x(a), \ldots, x(b))$ an aggregate vector consisting of $x(t)$, $t \in \mathbb{N}_{[a,b]}$, representing

a trajectory. When it is clear from the context, we write $\boldsymbol{x}(t)$, omitting the endpoint. When $x(t)$, $t \in \mathbb{N}_{[a,b]}$, are random vectors, $\boldsymbol{x}(a:b) = (x(a), \ldots, x(b))$ is a random process. Let $x_i(t)$, for $t \in \mathbb{N}_{[0,N]}$ and $i \in \mathbb{N}_{[1,M]}$. Then, $\boldsymbol{x}(0:N) = (x(0), \ldots, x(N))$ denotes an aggregate trajectory when $x(t) = (x_1(t), \ldots, x_M(t))$, $t \in \mathbb{N}_{[0,N]}$. The remainder of the division of a by b is $\mathrm{mod}(a,b)$. A random variable (vector) w following a distribution $\mathcal{D}_w$ is denoted as $w \sim \mathcal{D}_w$, the support of $\mathcal{D}_w$ is $\mathrm{supp}(\mathcal{D}_w)$, the expected value of w is $\mathbb{E}(w)$, and the variance (covariance matrix) of w is $\mathrm{Var}(w)$ ($\mathrm{Cov}(w)$). The probability of event Y is $\Pr\{Y\}$. The cardinality of a set $\mathcal{V}$ is $|\mathcal{V}|$. The Minkowski sum and the Pontryagin set difference of $S_1 \subseteq \mathbb{R}^n$ and $S_2 \subseteq \mathbb{R}^n$ are $S_1 \oplus S_2 = \{s_1 + s_2 \mid s_1 \in S_1,\ s_2 \in S_2\}$ and $S_1 \ominus S_2 = \{s_1 \in S_1 \mid s_1 + s_2 \in S_1, \forall s_2 \in S_2\}$, respectively.

Lemma 1 (Distributivity of Minkowski sum)**.** *Let $\mathcal{X}_i, \mathcal{Y}_i \subseteq \mathbb{R}^{n_i}$, $i \in \mathbb{N}_{[1,M]}$. Then, $(\mathcal{X}_1 \times \cdots \times \mathcal{X}_M) \oplus (\mathcal{Y}_1 \times \cdots \times \mathcal{Y}_M) = (\mathcal{X}_1 \oplus \mathcal{Y}_1) \times \cdots \times (\mathcal{X}_M \oplus \mathcal{Y}_M)$.*

As introduced in the previous chapter, we consider STL formulas with standard syntax

$$\varphi := \top \mid \pi \mid \neg\phi \mid \phi_1 \wedge \phi_2 \mid \phi_1 U_{[t_1,t_2]} \phi_2, \tag{3.1}$$

where $\pi := (\mu(x) \geq 0)$ is a predicate, $\mu(x) := a^\intercal x + b$ is an affine predicate function, with $a \in \mathbb{R}^{n_x}$, $x \in \mathbb{R}^{n_x}$, and $b \in \mathbb{R}$, and ϕ, ϕ_1, and ϕ_2 are STL formulas, which are built recursively using predicates π, logical operators $\neg$ and $\wedge$, and the *until* temporal operator U, with $[t_1, t_2] \equiv \mathbb{N}_{[t_1,t_2]}$. We omit $\vee$ (*or*), $\Diamond$ (*eventually*) and $\Box$ (*always*) operators from (3.1) and the sequel, as these may be defined by (3.1), e.g., $\phi_1 \vee \phi_2 = \neg(\neg\phi_1 \wedge \neg\phi_2)$, $\Diamond_{[t_1,t_2]}\phi = \top U_{[t_1,t_2]}\phi$, and $\Box_{[t_1,t_2]}\phi = \neg\Diamond_{[t_1,t_2]}\neg\phi$.

Let π be a predicate and ϕ an STL formula. We write $\pi \in \phi$ to indicate that π is part of the formula ϕ. We denote by $\boldsymbol{x}(t) \models \phi$, $t \in \mathbb{N}$, the satisfaction of ϕ, verified over $\boldsymbol{x}(t) = (x(t), x(t+1), \ldots)$. The validity of a formula ϕ can be verified using Boolean semantics: $\boldsymbol{x}(t) \models \pi \Leftrightarrow \mu(x(t)) \geq 0$, $\boldsymbol{x}(t) \models \neg\phi \Leftrightarrow \neg(\boldsymbol{x}(t) \models \phi)$, $\boldsymbol{x}(t) \models \phi_1 \wedge \phi_2 \Leftrightarrow \boldsymbol{x}(t) \models \phi_1 \wedge \boldsymbol{x}(t) \models \phi_2$, $\boldsymbol{x}(t) \models \phi_1 U_{[a,b]} \phi_2 \Leftrightarrow \exists \tau \in t \oplus \mathbb{N}_{[a,b]}$, s.t. $\boldsymbol{x}(\tau) \models \phi_2 \wedge \forall \tau' \in \mathbb{N}_{[t,\tau]}, \boldsymbol{x}(\tau') \models \phi_1$. Based on the Boolean semantics, the horizon of a formula is recursively defined as [14]: $N^\pi = 0$, $N^{\neg\phi} = N^\phi$, $N^{\phi_1 \wedge \phi_2} = \max(N^{\phi_1}, N^{\phi_2})$, $N^{\phi_1\, U_{[a,b]} \phi_2} = b + \max(N^{\phi_1}, N^{\phi_2})$.

STL is endowed with quantitative semantics [15]: A scalar-valued function $\rho^\phi : \mathbb{R}^n \times \cdots \times \mathbb{R}^n \to \mathbb{R}$ of a signal indicates how robustly a signal $\boldsymbol{x}(t)$ satisfies a formula ϕ. The robustness function is defined recursively as follows: $\rho^\pi(\boldsymbol{x}(t)) = \mu(x(t))$, $\rho^{\neg\phi}(\boldsymbol{x}(t)) = -\rho^\phi(\boldsymbol{x}(t))$, $\rho^{\phi_1 \wedge \phi_2}(\boldsymbol{x}(t)) = \min(\rho^{\phi_1}(\boldsymbol{x}(t)), \rho^{\phi_2}(\boldsymbol{x}(t)))$, and $\rho^{\phi_1 U_{[a,b]} \phi_2}(\boldsymbol{x}(t)) = \max_{\tau \in t \oplus \mathbb{N}_{[a,b]}} (\min(Y_1(\tau), Y_2(\tau')))$, with $Y_1(\tau) = \rho^{\phi_1}(\boldsymbol{x}(\tau))$, $Y_2(\tau') = \min_{\tau' \in \mathbb{N}_{[t,\tau]}} \rho^{\phi_2}(\boldsymbol{x}(\tau'))$, π being a predicate, and ϕ, ϕ_1, and ϕ_2 being STL formulas.

Definition 1. *Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be an undirected graph containing no self-loops, with node set $\mathcal{V}$, cardinality $M = |\mathcal{V}|$, and edge set $\mathcal{E}$. Let also $\mathcal{V}' \subseteq \mathcal{V}$, with $|\mathcal{V}'| > 1$, and define $\mathcal{E}_{\mathcal{V}'} \subseteq \mathcal{E}$ as the set of edges attached to nodes $\mathcal{V}'$. Then, $\mathcal{G}' = (\mathcal{V}', \mathcal{E}_{\mathcal{V}'})$ is a clique [16], i.e., a complete subgraph of $\mathcal{G}$, if $\mathcal{E}_{\mathcal{V}'}$ contains all possible edges between nodes $\mathcal{V}'$. The set of cliques of $\mathcal{G}$ is defined as $\mathcal{K} = \{\nu \subseteq \mathcal{V} \mid (\nu, \mathcal{E}_\nu)$ is a complete subgraph of $\mathcal{G}\}$.*

Consider a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with clique set $\mathcal{K}$, a clique $\nu \in \mathcal{K}$, with $\nu = (i_1, \ldots, i_{|\nu|})$, and vectors $x_{i_j}(t)$, $j \in \mathbb{N}_{[1,|\nu|]}$, with $t \in \mathbb{N}$, Then, $x_\nu(t) = (x_{i_1}(t), \ldots, x_{i_{|\nu|}}(t))$ is an aggregate vector. We denote by $\boldsymbol{x}_\nu(t) \models \phi_\nu$ the validity of an STL formula defined over the aggregate trajectory $\boldsymbol{x}_\nu(t) = (x_\nu(t), x_\nu(t+1), \ldots)$. If $\pi_\nu \in \phi_\nu$, $\pi_\nu := (\mu_\nu(x_\nu) \geq 0)$, where $\mu_\nu(x_\nu)$ is an affine predicate function of x_ν, with $x_\nu = (x_{i_1}, \ldots, x_{i_{|\nu|}})$.

3.2.2 Multi-agent system

Dynamics

We consider M agents with dynamics

$$x_i(t+1) = A_i x_i(t) + B_i u_i(t) + w_i(t), \tag{3.2}$$

where $x_i(t) \in \mathcal{X}_i \subseteq \mathbb{R}^{n_i}$, $u_i \in \mathcal{U}_i \subseteq \mathbb{R}^{m_i}$, and $w_i(t) \in \mathcal{W}_i \subseteq \mathbb{R}^{n_i}$ are the state, input and disturbance vectors, respectively, the initial condition, $x_i(0)$, is known, (A_i, B_i) is a stabilizable pair, with $A_i \in \mathbb{R}^{n_i \times n_i}$, $B_i \in \mathbb{R}^{n_i \times m_i}$, $i \in \mathbb{N}_{[1,M]}$, and $t \in \mathbb{N}$. By collecting individual state, input, and disturbance vectors, as $x(t) = (x_1(t), \ldots, x_M(t)) \in \mathcal{X} \subseteq \mathbb{R}^n$, $u(t) = (u_1(t), \ldots, u_M(t)) \in \mathcal{U} \subseteq \mathbb{R}^m$, and $w(t) = (w_1(t), \ldots, w_M(t)) \in \mathcal{W} \subseteq \mathbb{R}^n$, respectively, we write the dynamics of the entire MAS as

$$x(t+1) = Ax(t) + Bu(t) + w(t), \tag{3.3}$$

where $A = \mathrm{diag}(A_1, \ldots, A_M)$, $B = \mathrm{diag}(B_1, \ldots, B_M)$, and the state, input, and disturbance sets are $\mathcal{X} = \mathcal{X}_1 \times \cdots \times \mathcal{X}_M$, $\mathcal{U} = \mathcal{U}_1 \times \cdots \times \mathcal{U}_M$, and $\mathcal{W} = \mathcal{W}_1 \times \cdots \times \mathcal{W}_M$, respectively. We note that linearity is crucial for decomposing the MAS dynamics into deterministic and stochastic components, as discussed in Sec. 3.2.4. Such an approach can be extended to more general MAS with nonlinear dynamics, where the noise term is additive in the difference equation, such as $x(t+1) = f(x(t), u(t)) + g(w(t))$.

Disturbance

We assume that the uncertain sequence $\boldsymbol{w}_i(0) = (w_i(0), w_i(1), \ldots)$, with $i \in \mathbb{N}_{[1,M]}$, is an independent and identically distributed random process, and that $w_i(t)$ is a random vector with mean $\mathbb{E}(w_i(t)) = 0$ and positive definite covariance matrix $\mathrm{Cov}(w_i(t)) = Q_i$, which is known, for all $t \in \mathbb{N}$. We also assume that $w_i(t)$, $i \in \mathbb{N}_{[1,M]}$, are independent for $t \in \mathbb{N}$. We denote by $\mathcal{D}_{w_i}$ the distribution of the disturbance $w_i(t) \in \mathcal{W}_i$, where $\mathcal{W}_i$ is its support, which may be unbounded. We also may write that $w(t) \sim \mathcal{D}_w$, with $\mathrm{supp}(\mathcal{D}_w) = \mathcal{W} = \mathcal{W}_1 \times \cdots \times \mathcal{W}_M$, $\mathbb{E}(w(t)) = 0$, $\mathrm{Cov}(w(t)) = \mathrm{diag}(Q_1, \ldots, Q_M) = Q$.

STL specification

Let $\mathcal{V} = \{1, \ldots, M\}$ be the set collecting the indices of all agents. The MAS is subject to a conjunctive STL formula ϕ with syntax as in (3.1), where each conjunct is either a local subformula ϕ_i involving agent $i \in \mathcal{V}$, or a joint subformula ϕ_ν involving a subset of agents $\nu \subseteq \mathcal{V}$, where ν is a clique. By collecting all cliques ν in $\mathcal{K}_\phi$, the global STL task is

$$\phi = \bigwedge_{i \in \mathcal{V}} \phi_i \wedge \bigwedge_{\nu \in \mathcal{K}_\phi} \phi_\nu. \tag{3.4}$$

The structure of ϕ in (3.4) induces an interaction graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the set of nodes, and $\mathcal{E} = \{(\nu_i, \nu_j) \mid \nu_i, \nu_j \in \nu,\ i \neq j,\ \nu \in \mathcal{K}_\phi\}$ is the set of edges. Let $\pi := (\mu(y) \geq 0)$ be a predicate in ϕ, where $\mu(y) = a^\intercal y + b$, with $a, y \in \mathbb{R}^{n_y}$ and $b \in \mathbb{R}$. The vector $y \in \mathbb{R}^{n_y}$ represents either an individual state vector, $x_i \in \mathbb{R}^{n_i}$, $i \in \mathbb{N}_{[1,M]}$, or an aggregate vector, $x_\nu \in \mathbb{R}^{n_\nu}$, collecting the states of agents in the clique $\nu \in \mathcal{K}_\phi$. Formula ϕ can specify tasks between subsets of agents, by representing their entirety as cliques.

3.2.3 Problem statement

We wish to solve the stochastic optimal control problem

$$\underset{\boldsymbol{u}(0),\boldsymbol{x}(0)}{\text{Minimize}}\, \mathbb{E}\left[\sum_{i=1}^{M}\Big(\sum_{t=0}^{N-1}(\ell_i(x_i(t),u_i(t)))+V_{f,i}(x_i(N))\Big)\right] \tag{3.5a}$$

$$\text{s.t. } x(t+1)=Ax(t)+Bu(t)+w(t),\ t\in\mathbb{N}_{[0,N)}, \tag{3.5b}$$

$$\Pr\{\boldsymbol{x}(0)\models\phi\}\geq\theta,\ \text{with } x(0)=x_0, \tag{3.5c}$$

where $\ell_i:\mathbb{R}^{n_i}\times\mathbb{R}^{m_i}\to\mathbb{R}$, $V_{f,i}:\mathbb{R}^{n_i}\to\mathbb{R}$, the optimization variables are $\boldsymbol{u}(0)=(u(0),\dots,u(N-1))$, $\boldsymbol{x}(0)=(x(0),\dots,x(N))$, ϕ is a multi-agent STL formula, with structure as in (3.4) and syntax as in (3.1), to be satisfied by $\boldsymbol{x}(0)$ with a probability $\theta\in(0,1)$, x_0 is a known initial condition of the MAS, and N is the horizon of ϕ. Solving the problem directly is challenging due to the probabilistic constraint, the expectation operator in the cost function, and uncertain dynamics. To handle complexity, especially for a large number of agents and complex ϕ, we relax it with a deterministic problem, which subsequently, we decompose into smaller agent-level subproblems. Additionally, we make the following assumption.

Assumption 1. *For $x(0)=x_0$ and given $\theta\in(0,1)$, Problem (3.5) is feasible.*

3.2.4 Error dynamics and construction of probabilistic tubes

Due to the linearity in (3.2), the state of each agent can be decomposed into a deterministic part, $z_i(t)$, and an error, $e_i(t)$, i.e., $x_i(t)=z_i(t)+e_i(t)$. Consider the causal control law $u_i(t)=K_ie_i(t)+v_i(t)$, where $K_i\in\mathbb{R}^{m_i\times n_i}$ is a stabilizing state-feedback gain for the pair (A_i,B_i). Then,

$$z_i(t+1)=A_iz_i(t)+B_iv_i(t), \tag{3.6a}$$

$$e_i(t+1)=\bar{A}_ie_i(t)+w_i(t), \tag{3.6b}$$

where $z_i(0)=x_i(0)$, $e_i(0)=0$, and $\bar{A}_i=A_i+B_iK_i$. The above choice of $u_i(t)$ will allow us to control the size of the reachable sets of (3.6b), in light of the probabilistic constraint in (3.5c). Define now the aggregate vectors $z(t)=(z_1(t),\dots,z_M(t))$, $e(t)=(e_1(t),\dots,e_M(t))$, and $v(t)=(v_1(t),\dots,v_M(t))$, the block-diagonal state-feedback gain $K=\text{diag}(K_1,\dots,K_M)\in\mathbb{R}^{m\times n}$, and the block-diagonal closed-loop matrix $\bar{A}=\text{diag}(\bar{A}_1,\dots,\bar{A}_M)$. Then, we decompose (3.3) into

$$z(t+1)=Az(t)+Bv(t), \tag{3.7a}$$

$$e(t+1)=\bar{A}e(t)+w(t). \tag{3.7b}$$

Given a particular state feedback gain K, the error system (3.7b) can be analyzed independently of (3.7a). As a closed-loop system driven by the random vector $w(t)$, we can predict its trajectory $\boldsymbol{e}(0)=(e(0),\dots,e(N))$, with $e(0)=0$, by calculating its reachable sets probabilistically. Probabilistic reachable sets and tubes for system (3.7b) are defined next.

Definition 2. *A set $E(t)\subseteq\mathbb{R}^n$, $t\in\mathbb{N}_{[0,N]}$, is called a t-step probabilistic reachable set (t-PRS) for (3.7b) at probability level $\hat{\theta}_t\in[0,1]$, if $\Pr\{e(t)\in E(t)\mid e(0)=0\}\geq\hat{\theta}_t$.*

It is worth noting that for a probability level $\hat{\theta}_t$ a t-PRS $E(t)$, $t\in\mathbb{N}_{[0,N]}$, for (3.7b) is not unique.

Definition 3. *Let $\boldsymbol{e}(0) = (e(0), \dots, e(N))$ be a trajectory of (3.7b). Then, $\boldsymbol{E} \subseteq \mathbb{R}^n \times \cdots \times \mathbb{R}^n$ is called a probabilistic reachable tube (PRT) for (3.7b) at probability level $\Theta \in [0,1]$, if $\Pr\{\boldsymbol{e}(0) \in \boldsymbol{E}\} \geq \Theta$.*

Definition 4. *Let $w \sim \mathcal{D}_w$, with $\mathrm{supp}(\mathcal{D}_w) = \mathcal{W}$. We call $\mathscr{E}_\theta(\mathcal{D}_w) \subseteq \mathcal{W}$ a confidence region (CR) for $w \in \mathcal{W}$ at probability level θ, if $\Pr\{w \in \mathscr{E}_\theta(\mathcal{D}_w)\} \geq \theta$.*

CRs for $w_i(t)$, $i \in \mathbb{N}_{[1,M]}$, can be approximated via Monte Carlo methods or computed analytically using concentration inequalities depending on the properties of $\mathcal{D}_w$. Here, since $\mathbb{E}(w_i(t)) = 0$ and $\mathrm{Cov}(w_i(t)) = Q_i > 0$, for $t \in \mathbb{N}$ and $i \in \mathbb{N}_{[1,M]}$, we construct ellipsoidal CRs at probability level θ_i as $\mathscr{E}_{\theta_i}(\mathcal{D}_{w_i}) = \{w_i \in \mathbb{R}^{n_i} \mid w_i^\intercal Q_i^{-1} w_i \leq n_i/\theta_i\}$, by the multivariate Chebyshev's inequality. Next, we construct a CR for the aggregate random vector $w(t) = (w_1(t), \dots, w_M(t))$.

Lemma 2. *Let $\mathscr{E}_{\theta_i}(\mathcal{D}_{w_i})$ be a CR for $w_i(t) \in \mathcal{W}_i$, $i \in \mathbb{N}_{[1,M]}$, at probability level θ_i, for all $t \in \mathbb{N}$. Then, $\mathscr{E}_{\hat{\theta}}(\mathcal{D}_w) = \mathscr{E}_{\theta_1}(\mathcal{D}_{w_1}) \times \cdots \times \mathscr{E}_{\theta_M}(\mathcal{D}_{w_M})$, is a CR of $w(t) \in \mathcal{W}$ at probability level $\hat{\theta}$ for all $t \in \mathbb{N}$, where $\hat{\theta} \geq \Pi_{i=1}^M \theta_i$.*

Based on the CR construction of the disturbance $w(t)$, we construct t-PRSs at certain probability levels for the multi-agent error system (3.7b) as follows.

Proposition 1. *Let $\mathscr{E}_{\hat{\theta}}(\mathcal{D}_w) = \mathscr{E}_{\theta_1}(\mathcal{D}_{w_1}) \times \cdots \times \mathscr{E}_{\theta_M}(\mathcal{D}_{w_M})$ be a CR for $w(t)$, where $\mathscr{E}_{\theta_i}(\mathcal{D}_{w_i})$ is an ellipsoidal CR for $w_i(t)$ at probability level θ_i, $i \in \mathbb{N}_{[1,M]}$. Then, the sets $E(t) \subseteq \mathbb{R}^n$, $t \in \mathbb{N}_{[0,N]}$, which are recursively defined as $E(t+1) = \bar{A}E(t) \oplus \mathscr{E}_{\hat{\theta}}(\mathcal{D}_w)$, with $E(0) = \{0\} \times \cdots \times \{0\}$, are t-PRSs for (3.7b) at probability level $\hat{\theta} \geq \Pi_{i=1}^M \theta_i$, and $E(t) = E_1(t) \times \cdots \times E_M(t)$, $t \in \mathbb{N}_{[0,N]}$, where $E_i(t)$ is a t-PRS for (3.6b) at probability level θ_i, with $i \in \mathbb{N}_{[1,M]}$.*

Prop. 1 leads to the following PRT result.

Theorem 1. *Let $\mathscr{E}_{\hat{\theta}}(\mathcal{D}_w) = \mathscr{E}_{\theta_1}(\mathcal{D}_{w_1}) \times \cdots \times \mathscr{E}_{\theta_M}(\mathcal{D}_{w_M})$ be a CR for $w(t)$, and define t-PRSs, $E(t)$, $t \in \mathbb{N}_{[0,N]}$ for system (3.7b) at probability level $\hat{\theta} \geq \Pi_{i=1}^M \theta_i$, where θ_i is the confidence level of the region $\mathscr{E}_{\theta_i}(\mathcal{D}_{w_i})$, $i \in \mathbb{N}_{[1,M]}$, as in Prop. 1. Then, i) $\boldsymbol{E}_i = E_i(0) \times \cdots \times E_i(N)$ is a PRT for (3.6b) at probability level $\Theta_i \geq 1 - N(1-\theta_i)$, where $E_i(t)$, $t \in \mathbb{N}_{[0,N]}$, is a t-PRS for (3.6b) at probability level θ_i. ii) $\boldsymbol{E} = E(0) \times \cdots \times E(N)$ is a PRT for (3.7b) at probability level $\Theta = \Pi_{i=1}^M \Theta_i$. iii) Let $e_\nu(t+1) = \bar{A}_\nu e_\nu(t) + w_\nu(t)$ be the aggregate system collecting individual error systems from the clique $\nu \in \mathcal{K}_\phi$, where $\nu = (i_1, \dots, i_{|\nu|})$, $e_\nu(t) = (e_{i_1}(t), \dots, e_{i_{|\nu|}}(t))$, $w_\nu(t) = (w_{i_1}(t), \dots, w_{i_{|\nu|}}(t))$, and $\bar{A}_\nu = \mathrm{diag}(\bar{A}_{i_1}, \dots, \bar{A}_{i_{|\nu|}})$, and let $E_\nu(t) = E_{i_1}(t) \times \cdots \times E_{i_{|\nu|}}(t)$, $t \in \mathbb{N}_{[0,N]}$, be its t-PRSs, with $E_{i_j}(t)$, being t-PRS for (3.6b) at probability level θ_{i_j}, with $j \in \mathbb{N}_{[1,|\nu|]}$. Then, $\boldsymbol{E}_\nu = E_\nu(0) \times \cdots \times E_\nu(N)$ is a PRT at probability level $\Theta = \Pi_{j=1}^{|\nu|} \Theta_{i_j}$.*

Remark 1. *Note that our PRT construction reduces conservatism for a large number of agents, while utilizing the union-bound argument only over time. This may require conservative choices for the probability levels, θ_i, for the CRs of $w_i(t)$, $i \in \mathbb{N}_{[1,M]}$, for large horizons. To construct, e.g., a PRT for (3.7b) at probability level Θ, one may select uniform probability levels for the CRs as $\theta_i \geq 1 - \frac{1-\Theta^{\frac{1}{M}}}{N}$, where $\theta_i \to 1$ for large N, regardless of Θ.*

3.2.5 Constraint tightening

We aim to design a trajectory for the deterministic system (3.7a) that satisfies an STL formula derived from ϕ, incorporating *tighter* predicates. The following proposition underpins this approach.

Proposition 2. *Let $\boldsymbol{x}(0) = \boldsymbol{z}(0) + \boldsymbol{e}(0)$, with $\boldsymbol{x}(0) = (x(0), \dots, x(N))$, $\boldsymbol{z}(0) = (z(0), \dots, z(N))$ and $\boldsymbol{e}(0) = (e(0), \dots, e(N))$. Suppose that $\Pr\{\boldsymbol{e}(0) \in \boldsymbol{E}\} \geq \theta$, for some $\boldsymbol{E} = E(0) \times \cdots \times E(N)$, with $E(t) \subseteq \mathbb{R}^n$, $t \in \mathbb{N}$. If $\boldsymbol{z}(0) + \boldsymbol{e}(0) \models \phi$ for all $\boldsymbol{e}(0) \in \boldsymbol{E}$, then $\Pr\{\boldsymbol{x}(0) \models \phi\} \geq \theta$.*

Let $\boldsymbol{E}$ be a PRT for (3.7b) at probability level Θ. Next, we construct a formula ψ such that $\boldsymbol{z}(0) \models \psi$ implies that $\boldsymbol{z}(0) + \boldsymbol{e}(0) \models \phi$, for all $\boldsymbol{e}(0) \in \boldsymbol{E}$, that is, $\Pr\{\boldsymbol{x}(0) \models \phi\} \geq \theta$, by Prop. 2. Formula ψ has identical Boolean and temporal operators with ϕ in (3.4), and retains its multi-agent structure:

$$\psi = \bigwedge_{i \in \mathcal{V}} \psi_i \wedge \bigwedge_{\nu \in \mathcal{K}_\phi} \psi_\nu. \tag{3.8}$$

Let $\pi := (a^\intercal y + b \geq 0)$ be a predicate, with $a, y \in \mathbb{R}^{n_y}$, $b \in \mathbb{R}$. We denote by $\tau(\pi)$ the *tighter* version of π, where $\tau(\pi) \in \psi$ if $\pi \in \phi$, and $\neg\tau(\pi) \in \psi$ if $\neg\pi \in \phi$, with

$$\tau(\pi) := (a^\intercal y + b + \min_{g \in G} a^\intercal g \geq 0), \text{ if } \tau(\pi) \in \psi, \tag{3.9a}$$

$$\tau(\pi) := (a^\intercal y + b + \max_{g \in G} a^\intercal g \geq 0), \text{if } \neg\tau(\pi) \in \psi. \tag{3.9b}$$

Here, $G = \bigcup_{t=1}^{N} E_i(t)$ if $y = x_i(t)$, for some $i \in \mathcal{V}$, or $G = \bigcup_{t=1}^{N} E_\nu(t)$ if $y = x_\nu(t)$ for some $\nu \in \mathcal{K}_\phi$. We remark that the optimizations in (3.9) are tractable since the domain G is the union of finitely many convex and compact sets by the construction of t-PRSs, $E_i(t)$, $E_\nu(t)$, $t \in \mathbb{N}_{[1,N]}$, in Prop. 1 and Thm. 1. Practically, the tightening in (3.9) can be retrieved by solving a convex optimization by taking the convex hull of G, or, better, by solving N convex optimization problems, one for every $E_i(t)$, $E_\nu(t)$, $t \in \mathbb{N}_{[1,N]}$, and selecting the worst-case (minimum for (3.9a) and maximum for (3.9b)) solution among them. We are now ready to state the following result.

Theorem 2. *Let ψ be the STL formula resulting from ϕ according to (3.8)-(3.9), and assume that the deterministic problem:*

$$\underset{\boldsymbol{v}(0), \boldsymbol{z}(0)}{\text{Minimize}} \sum_{i=1}^{M} \left(\sum_{t=0}^{N-1} (\ell_i(z_i(t), v_i(t))) + V_{f,i}(z_i(N)) \right) \tag{3.10a}$$

$$\text{s.t. } z(t+1) = Az(t) + Bv(t), \ t \in \mathbb{N}_{[0,N)}, \tag{3.10b}$$

$$\boldsymbol{z}(0) \models \psi, \text{ with } z(0) = x_0, \tag{3.10c}$$

has a feasible solution $\boldsymbol{v}(0) = (v(0), \dots, v(N-1))$, with $v(t) \in \mathcal{U} \ominus KE(t)$, $t \in \mathbb{N}_{[0,N-1]}$, where K is a stabilising gain for (A, B) in (3.7b), and $E(t) = E_1(t) \times \cdots \times E_M(t)$, $E_i(t)$, $i \in \mathbb{N}_{[1,M]}$, being t-PRS for (3.6b), at probability level θ_i, such that $\Pi_{i=1}^{M} (1 - N(1 - \theta_i)) \geq \theta$. Let $\boldsymbol{e}(0) = (e(0), \dots, e(N))$ be a trajectory of (3.7b). Then, $\boldsymbol{u}(0) = \text{diag}(K, \dots, K)\boldsymbol{e}(0) + \boldsymbol{v}(0)$ is a feasible solution for (3.5).

3.2.6 Distributed control synthesis

We propose an iterative procedure that handles the complexity of (3.10). First, we assume the following.

Assumption 2. *The optimization (3.10) has a feasible solution* $\boldsymbol{v}(0) = (v(0), \ldots, v(N-1))$, $\boldsymbol{z}(0) = (z(0), \ldots, z(N))$, *where* $v(t) \in \mathcal{U} \ominus KE(t)$, $t \in \mathbb{N}_{[0,N)}$, *with the gain* K *and the t-PRSs* $E(t)$, $t \in \mathbb{N}_{[0,N]}$, *being as in Thm. 2.*

Decomposition of STL formula ψ

For a node i participating in at least one clique, i.e., $i \in \nu$, with $\nu \in \mathcal{K}_\phi$, we define $\mathcal{T}_i$ by the set of cliques containing i excluding i, i.e.,

$$\mathcal{T}_i = \{\nu \setminus i : \nu \in \mathrm{cl}(i)\}, \tag{3.11}$$

where $\mathrm{cl}(i) = \{\nu \in \mathcal{K}_\phi,\ \nu \ni i\}$ is the set of cliques that contain i. Let $j \in \mathcal{T}_i$, with $j = (i_1, \ldots, i_{|j|})$. Let a trajectory $\boldsymbol{z}_{ij}(0) = (z_{ij}(0), \ldots, z_{ij}(N))$, where $z_{ij}(t) = (z_{i_1}(t), \ldots, z_i(t), \ldots, z_{i_{|j|}}(t))$, with $t \in \mathbb{N}_{[0,N]}$, and the order $i_1 < \ldots < i < \ldots < i_{|j|}$ being specified by the lexicographic ordering of the node set $\mathcal{V} = \mathbb{N}_{[1,M]}$. Using (3.11), an equivalent formula to the *tighter* formula (3.8)-(3.9), ψ, is defined as $\hat{\psi} = \bigwedge_{i \in \mathcal{V}} \hat{\psi}_i$, where

$$\hat{\psi}_i = \psi_i \wedge \bigwedge_{j \in \mathcal{T}_i} \psi_{ij}. \tag{3.12}$$

Iterative algorithm

For simplicity, we drop the time argument and introduce an iteration index as a superscript in the trajectory notation, e.g., $\boldsymbol{z}_i^k$ ($\boldsymbol{z}_{ij}^k$) indicates a trajectory $\boldsymbol{z}_i(0)$ ($\boldsymbol{z}_{ij}(0)$) that is retrieved at the kth iteration of the following procedure. To initialize the procedure, we generate initial guesses on the agents' trajectories by solving

$$\underset{\boldsymbol{v}_i^0, \boldsymbol{z}_i^0}{\text{Minimize}} \sum_{t=0}^{N-1} (\ell_i(z_i^0(t), v_i^0(t))) + V_{f,i}(z_i^0(N)) \tag{3.13a}$$

$$\text{s.t. } z_i^0(t+1) = A_i z_i^0(t) + B_i v_i^0(t),\ t \in \mathbb{N}_{[0,N)}, \tag{3.13b}$$

$$\boldsymbol{z}_i^0 \models \psi_i, \text{ with } z_i^0(0) = x_{0,i}, \tag{3.13c}$$

at $k = 0$ for $i \in \mathbb{N}_{[1,M]}$. After solving problem (3.13), which is feasible by Assumption 2, at iteration $k \geq 1$, only a subset of agents, denoted by $\mathcal{O}_k \subset \mathcal{V}$, are allowed to update their input sequences by performing an optimization. The remaining agents retrieve their input sequences from the previous iteration $k-1 \geq 0$. Roughly, the set $\mathcal{O}_k \subset \mathcal{V}$ is constructed so that any combination of its elements does not belong to a clique $\nu \in \mathcal{K}_\phi$. Due to space limitations, we simply construct $\mathcal{O}_k$ as a singleton, which only affects the number of agents' trajectories that can be optimized in parallel per iteration. For the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, with $\mathcal{V} = \mathbb{N}_{[1,M]}$, $\mathcal{O}_k = \mathrm{mod}(\mathcal{O}_{k-1}, M) + 1$, for $k > 1$, with $\mathcal{O}_1 = 1$. We refer readers to [17, Sec. V.B] for a more efficient construction of $\mathcal{O}_k$ enabling parallel computations at each iteration (see Sec. 3.2.7 for a numerical example).

At the kth iteration, with $k \geq 1$, if $i \notin \mathcal{O}_k$, then, $\boldsymbol{z}_i^k = \boldsymbol{z}_i^{k-1}$ and $\boldsymbol{v}_i^k = \boldsymbol{v}_i^{k-1}$. Otherwise, the input sequence of the ith agent is updated by solving

$$\underset{\boldsymbol{v}_i^k, \boldsymbol{z}_i^k}{\text{Minimize}} \sum_{t=0}^{N-1} (\ell_i(z_i^k(t), v_i^k(t))) + V_{f,i}(z_i^k(N)) - \mu_{ij_k}^k \tag{3.14a}$$

$$\text{s.t. } z_i^k(t+1) = A_i z_i^k(t) + B_i v_i^k(t), \; t \in \mathbb{N}_{[0,N)}, \tag{3.14b}$$

$$\boldsymbol{z}_i^k \models \psi_i, \text{ with } z_i(0) = x_{0,i}, \tag{3.14c}$$

$$\rho^{\psi_{ij_k}}(\boldsymbol{z}_{ij_k}^k) \geq \mu_{ij_k}^k, \; j_k = \underset{j \in \mathcal{T}_i}{\operatorname{argmin}} \{\rho^{\psi_{ij}}(\boldsymbol{z}_{ij}^{k-1})\}, \tag{3.14d}$$

$$\mu_{ij_k}^k \geq \min\left(0, \rho^{\psi_{ij_k}}(\boldsymbol{z}_{ij_k}^{k-1})\right), \tag{3.14e}$$

$$\rho^{\psi_{ij}}(\boldsymbol{z}_{ij}^k) \geq \min\left(0, \rho^{\psi_{ij}}(\boldsymbol{z}_{ij}^{k-1})\right), \; \forall j \in \mathcal{T}_i \setminus j_k, \tag{3.14f}$$

where $\rho^{\psi_{ij}}(\boldsymbol{z}_{ij}^k)$ is the robustness function of the formula ψ_{ij} evaluated over the trajectory $\boldsymbol{z}_{ij}^k$. Agent-i, with $i \in \mathcal{O}_k$, by solving (3.14), retrieves an input sequence that guarantees 1) the satisfaction of the individual task ψ_i (see constraint (3.14c)), 2) the improvement of the most violating (or least robust) joint task ψ_{ij_k} (see constraints (3.14d)-(3.14e)), and 3) either improvement on or non-violation of the remaining joint tasks (see constraint (3.14f)). The inclusion of the min operator in the constraints (3.14e)-(3.14f) relaxes the satisfaction of joint tasks that have already been found to be satisfiable in previous iterations. This allows the algorithm to emphasize the satisfaction of joint tasks with the smallest robustness function. The algorithm may terminate if it exceeds a maximum number of iterations, denoted as $k_{\max}$ and defined by the designer, yielding a minimally violating solution. Alternatively, termination occurs when verifying the satisfiability of all joint tasks, i.e., when $\mu_{ij}^k \geq 0$ for all $i \in \mathcal{V}$, $j \in \mathcal{T}_i$, and some $k \leq k_{\max}$, returning a feasible solution to (3.10). The overall iterative procedure is summarized in Alg. 1, the integrity of which relies on the following result.

Theorem 3. *At each iteration $k \geq 1$, the optimization problem (3.14) is feasible for all $i \in \mathcal{O}_k$.*

Algorithm 1 Iterative procedure for solving (3.10)

```
Compute T_i (3.11) and construct ψ̂_i (3.12), for i ∈ N_[1,M]
Solve (3.13) and store v_i^0, z_i^0, for i ∈ N_[1,M]
Construct O_k, for k ∈ N_[1,k_max]
for k in 1 : k_max do
    for i in 1 : M do
        if i ∈ O_k, solve (3.14), and store (v_i^k, z_i^k)
        if i ∉ O_k, update v_i^k ← v_i^{k-1} and z_i^k ← z_i^{k-1}
    Construct (v(0), z(0)) from (v_i^k, z_i^k), i ∈ N_[1,M]
    if ρ^ψ(z(0)) ≥ 0 go to 10
return (v(0), z(0))
```

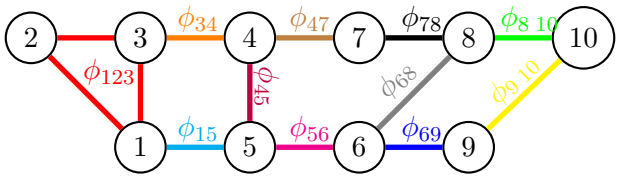

Figure 3.1: The ten agents of the MAS and the cliques $\mathcal{K}_\phi$ in ϕ.

3.2.7 Ten-Agent Example

We consider ten agents with aggregate dynamics given by $x(t+1) = x(t) + u(t) + w(t)$, where $x(t) = (x_1(t), ..., x_{10}(t)) \in \mathbb{R}^{20}$, $u(t) = (u_1(t), ..., u_{10}(t)) \in \mathbb{R}^{20}$, and $w(t) = (w_1(t), ..., w_{10}(t)) \in \mathbb{R}^{20}$. States $x_i(t) \in \mathcal{X}$, where $\mathcal{X}$ is the workspace confined by the dashed border in Fig. 3.2. Individual inputs are constrained by $\|u_i(t)\|_\infty \leq 0.8$, and disturbances, $w_i(t)$, are Gaussian random vectors, independent time- and agent-wise, with zero mean and covariance, $Q_i = 0.05 I_2$, for all $t \in \mathbb{N}$ and $i \in \mathbb{N}_{[1,10]}$. The MAS is assigned a specification $\phi = \bigwedge_{i=1}^{10} \phi_i \wedge \bigwedge_{\nu \in \mathcal{K}_\phi} \phi_\nu$, with horizon $N = 100$, where $\mathcal{K}_\phi$ is the set of cliques shown in Fig. 3.1, and ϕ_i, ϕ_ν, are tasks assigned to agent i, and the agents in $\nu \in \mathcal{K}_\phi$, respectively. In (3.5), we select $\ell_i(x_i(t), u_i(t)) = \|u_i(t)\|_1$, $V_{f,i}(x_i(100)) = 0$, for all $i \in \mathbb{N}_{[1,10]}$, and set $\theta = 0.70$.

Let $\phi_i = \left(\Box_{[0,100]}(\varphi_i^{\mathcal{X}} \wedge \neg\varphi_i^{O_1} \wedge \neg\varphi_i^{O_2} \wedge \neg\varphi_i^{O_3})\right) \wedge \left(\Diamond_{[10,50]}\varphi_i^{T_i}\right) \wedge \left(\Diamond_{[70,100]}\varphi_i^{G_i}\right)$ be an individual task, which requires agent-i, starting from $x_i(0)$ to pass through T_i and G_i within the intervals $\mathbb{N}_{[10,50]}$ and $\mathbb{N}_{[70,100]}$, respectively, while always staying within $\mathcal{X}$ and avoiding O_1, O_2, O_3. Regions T_i, G_i, $i \in \mathbb{N}_{[1,10]}$, and obstacles O_1, O_2, O_3, are in Fig. 3.2. Note that $x_i(t) \models \varphi_i^{\mathcal{Y}}$ if $x_i(t) \in \mathcal{Y}$, $\mathcal{Y} = \{\mathcal{X}, O_1, O_2, O_3, T_1, \ldots, T_{10}, G_1, \ldots, G_{10}\}$, for $t \in \mathbb{N}_{[0,100]}$.

Let $\phi_\nu = \Diamond_{[0,100]}\left(\|C_\nu x_\nu(t)\|_\infty \leq 1\right)$, where $C_\nu = \begin{bmatrix} I & -I \end{bmatrix}$ if $|\nu| = 2$ or $C_\nu = \begin{bmatrix} I & -I & 0 \\ 0 & I & -I \\ I & 0 & -I \end{bmatrix}$ if $|\nu| = 3$, be a joint task requiring agents in $\nu \in \mathcal{K}_\phi$ (see Fig. 3.1) to approach one another at least once within the horizon.

To formulate the deterministic problem (3.10), we proceed as follows: First, we select closed-loop matrices $\bar{A}_i = I_2 + K_i$, with $K_i = -0.5 I_2$, $i \in \mathbb{N}_{[1,10]}$, and construct t-PRSs, $E_i(t)$, $t \in \mathbb{N}_{[0,100]}$, for (3.6b), by the recursion $E_i(t+1) = \bar{A}_i E_i(t) \oplus \mathscr{E}_{\theta_i}(\mathcal{D}_{w_i})$, with $E_i(0) = \{0\}$, at probability levels $\theta_i = 1 - \frac{1-0.7^{\frac{1}{10}}}{100} = 0.9996$, $i \in \mathbb{N}_{[1,10]}$, such that $\Pi_{i=1}^{10}(1 - 100(1-\theta_i)) \geq 0.7$, and $\mathscr{E}_{\theta_i}(\mathcal{D}_{w_i}) = \{w_i | w_i^\intercal Q_i^{-1} w_i \leq \chi_2^2(\theta_i)\}$, where χ_2^2 is the chi-squared distribution of degree 2. Last, given that $\boldsymbol{E} = E(0) \times \cdots \times E(100)$, with $E(t) = E_1(t) \times \cdots \times E_{10}(t)$, $t \in \mathbb{N}_{[0,100]}$, is a PRT for (3.7b) at probability level $\Theta = 0.7$, by Thm. 1, we perform the optimizations in (3.9), derive the *tighter* formula ψ as in (3.8), and formulate (3.10) as an MILP. We have attempted to solve (3.10) in a centralized manner using the GUROBI solver [18], which produces a solution after running for 2.5 hours, with its feasibility iteration limit set to a maximum of ten. To obtain a solution faster, we first decompose ψ according to (3.12), based on the sets $\mathcal{T}_1 = \{(2,3),5\}$, $\mathcal{T}_2 = \{(1,3)\}$, $\mathcal{T}_3 = \{(1,2),4\}$, $\mathcal{T}_4 = \{3,5\}$, $\mathcal{T}_5 = \{1,4,6\}$, $\mathcal{T}_6 = \{5,8,9\}$, $\mathcal{T}_7 = \{4,8\}$, $\mathcal{T}_8 = \{6,7,10\}$, $\mathcal{T}_9 = \{6,10\}$, and $\mathcal{T}_{10} = \{8,9\}$. By selecting sets $\mathcal{O}_k$, $k \geq 1$, as $\mathcal{O}_1 = \{1,4,6,10\}$, $\mathcal{O}_2 = \{3,5,7,9\}$, $\mathcal{O}_3 = \{8,9,2,5\}$, $\mathcal{O}_4 = \mathcal{O}_1$, $\mathcal{O}_5 = \mathcal{O}_2$, $\mathcal{O}_6 = \mathcal{O}_3$, $\mathcal{O}_7 = \mathcal{O}_1$, and so on, we run Alg. 1, which terminates in less than six minutes returning a multi-agent trajectory, illustrated in Fig. 3.2, that satisfies the global STL task ψ.

Fig. 3.3 shows the computational overhead of Alg. 1 vs. the centralized solution for varying agent numbers, using a log scale to highlight the different runtime magnitudes. Note that the runtime of Alg. 1 can further be improved if agent-level subproblems, (3.13), (3.14), can be solved in parallel. By evaluating the robustness function of ϕ for numerous noisy realizations, we see that ϕ is violated in less than 30% of the time, satisfying the specification probability $\theta = 0.7$ and thereby verifying Thm. 2.

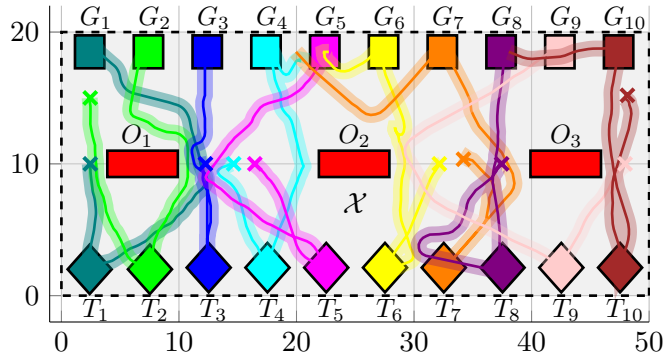

Figure 3.2: Nominal trajectories (solid lines) by Alg. 1, with initial states marked by crosses. Tubes (transparent covers) around trajectories, at probability levels $\Theta_i = 0.965$.

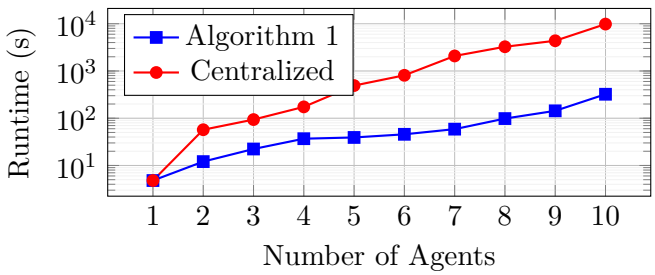

Figure 3.3: Compute times (log scale) for solving (3.10) via Alg. 1 and a centralized approach for varying agent numbers.

3.3 Distributionally Robust Optimization (DRO)

DRO is an approach to optimization under stochastic uncertainty in which the underlying true probability distribution is not available, and instead, one has to make decisions based on finitely many data observed from the system. This approach is effective even when the system dynamics are nonlinear, as it does not rely on the linearity of the dynamics.

Let us assume that a decision maker seeks to choose a decision variable u (e.g., control inputs) from a feasible set U (e.g., the admissible control set) to minimize a certain objective cost ℓ. Due to uncertainties, the objective function $\ell : U \times W \to \mathbb{R}$ depends on both the decision u and an uncertain quantity $w \in W$, and returns a real value. It is common to assume that w is a random variable with distribution P supported on W. Given this setup, the following stochastic optimization problem can be formulated to determine the decision:

$$\inf_{u \in U} \mathbb{E}_P[\ell(u, w)], \tag{3.15}$$

where $\mathbb{E}_P[\cdot]$ denotes the expected value under the probability distribution P.

In order to solve (3.15), one typically needs to know the underlying distribution P. In most practical situations, however, we only have access to N realizations $\{w_1, \ldots, w_N\}$ drawn from the underlying distribution. A common scenario-based approach is to use the following empirical risk minimization:

$$\inf_{u \in U} \frac{1}{N} \sum_{i=1}^{N} \ell(u, w_i), \tag{3.16}$$

as an approximation to (3.15). A key question that arises is how well the solution of (3.16) approximates the solution to (3.15). This is often referred to as the *out-of-sample guarantee*. Most guarantees related to (3.16) require certain assumptions on ℓ, such as convexity and boundedness [19]. Additionally, these out-of-sample guarantees tend to hold when the number of realizations N is sufficiently large.

Alternatively, DRO does not rely on knowing the true underlying distribution. In this approach, a set $\mathbb{P}$ of probability distributions, known as the *ambiguity set*, is defined, and the following minimax problem is solved:

$$\inf_{u \in U} \sup_{Q \in \mathbb{P}} \mathbb{E}_Q[\ell(u, w)]. \tag{3.17}$$

DRO offers robust guarantees on the out-of-sample performance, with a limited number of samples, by considering the worst-case distribution within an ambiguity set and adjusting the size of the ambiguity set such that $\mathbb{P}$ contains the true probability with a high confidence. Various methods exist for constructing ambiguity sets, such as moment ambiguity [20], Kullback-Leibler divergence-based ball [21], and Wasserstein-based ball [22]. The Wasserstein ambiguity set represents a statistical ball within the space of probability distributions surrounding the empirical distribution, with its radius measured using Wasserstein distance. Wasserstein DRO offers a probabilistic guarantee based on finite samples within a manageable formulation [23] and has attracted significant attention recently [24]. The Wasserstein-based ambiguity set (ball) is defined as (see Figure 3.4):

$$\mathbb{P} := \{Q \in \mathcal{Q}(W) \,|\, d_{\mathrm{W}}(Q, \hat{P}) \leq r\},$$

where $Q(W)$ denotes the set of Borel probability measures on the support W, $r \geq 0$ is the radius of the ball, $\hat{P}$ is the empirical distribution, defined as follows:

$$\hat{P} = \frac{1}{N} \sum_{i=1}^{N} \delta_{w_i},$$

where δ_{w_i} is the Dirac measure concentrated at w_i and $d_{\mathrm{W}} : Q(W) \times Q(W) \to \mathbb{R}_{\geq 0}$ is the Wasserstein metric, defined as follows:

$$d_{\mathrm{W}}(Q^1, Q^2) := \min_{\kappa \in Q(W^2)} \left\{ \int_{W^2} \|w_1 - w_2\| \mathrm{d}\kappa(w_1, w_2) \;\middle|\; \Pi^j \kappa = Q^j, j = 1, 2 \right\},$$

for all distributions $Q^1, Q^2 \in Q(W)$ where $\Pi^j \kappa$ denotes the jth marginal of the joint distribution κ for $j = 1, 2$.

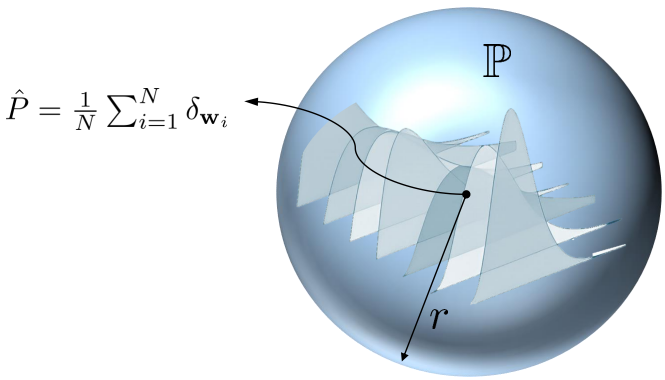

Figure 3.4: Wasserstein ambiguity ball: Set of distributions within a specified distance from a nominal one.

It has been shown that under mild assumptions, the distributionally robust optimization problems over Wasserstein balls can in fact be reformulated as finite convex programs–in many interesting cases even as tractable linear programs. In other words, the problem of determining the worst-case expectation (3.17) has desirable tractability properties.

3.3.1 STL Specifications using DRO

Task planning for multi-agent stochastic systems under STL specifications can be formulated as a Chance-Constrained Program (CCP). In the presence of process noise, the objective–such as minimizing a cost function in expectation–is optimized while probabilistically satisfying a given STL constraint using robustness function of the STL. For example, consider a robot tasked with reaching a target while avoiding collisions with another agent in the environment. The robot is affected by environmental disturbances and must also account for the other agent's autonomous dynamics, which may have inherent uncertainties. This scenario can be formulated as a robust CCP.

Figure 3.5 illustrates the interaction between a robot and an uncertain agent. The robot's planned path is represented by blue circles, corresponding to discrete time samples, while the car's

path is depicted by filled black circles. From the robot's perspective, the car is modeled as an autonomous agent with parametric uncertainties, shown in block diagrams. The objective is to generate a task plan that enables the robot to reach the green area while avoiding all possible car trajectories. In the scenario presented, there are three time instances, indicated by red, where a potential collision between the robot and the car may occur.

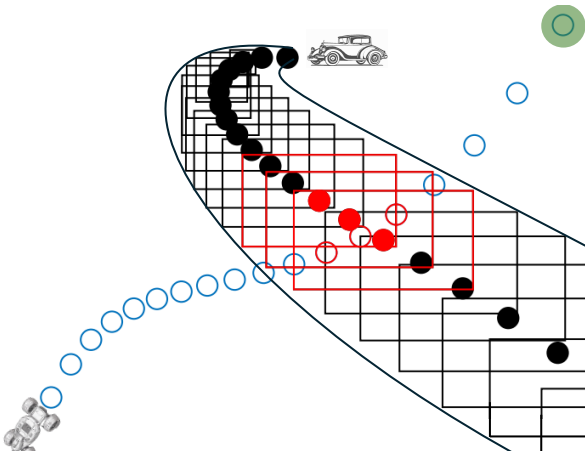

Figure 3.5: A robot is interacting with an uncertain agent and stochastic environment.

The CCP can then be transformed into a stochastic program with expected constraints, called Expected Constraint Program (ECP), using various methods, such as concentration of measure [25] and Conditional Value at Risk (CVaR) [26]. This transformation allows for a more tractable optimization problem where the goal is to balance minimizing the objective with satisfying the constraints in *expectation*. To achieve this, it is necessary to estimate the expected outcomes using a finite number of samples from the system's stochastic processes. As explained earlier, one effective approach for this is to apply DRO techniques.

Given that the robustness functions in STL are generally non-convex, developing an equivalent optimization approach that does not rely on convexity assumptions is crucial. Inspired by the results in [23] and [24], we have provided a data-driven equivalent solution to the DRO in [27]. We have guaranteed that any feasible solution derived from the resulting data-driven optimization is also a feasible solution to the original ECP (with the true distribution) with a predefined confidence level. Furthermore, the control inputs obtained from the proposed optimization offer performance guarantees with respect to the true distribution, with high confidence.

The robustness function in STL is used to ensure the satisfaction of the STL specification. Specifically, the robustness function is composed of several min and max operators, each corresponding to a sub-task of the STL. Satisfaction of the entire STL formula is guaranteed when the resulting robustness function is positive. However, the min and max operators used in defining the robustness function are not smooth, which poses challenges for numerical solvers that often struggle with non-smooth functions.

Inspired by [28], we can address this issue by using smooth under-approximations of the min and max operators. This can be achieved through the application of logarithmic and exponential functions with adjustable constants to control the accuracy of the approximation. By ensuring that the approximated robustness function is positive, we can guarantee that the exact robustness

function is also positive, and thus, that the STL specification is satisfied.

Figure 3.6 illustrates this concept using a simple example involving the min and max of three linear functions. The red and the green curves provide smooth under approximations of max and min of three blue lines, respectively.

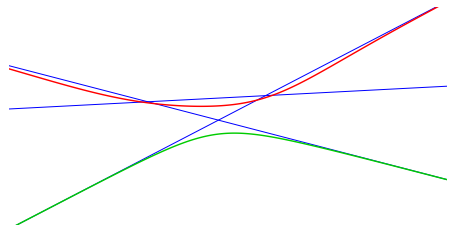

Figure 3.6: Smooth under-approximation of min and max operators.

Figure 3.7, shows an example of a two-dimensional linear stochastic system. The goal is to ensure the system adheres to a safety constraint for all time steps within a fixed horizon and reaches a specific target region with high probability, while also minimizing a quadratic cost function. As it can be seen, due to the robustness nature of the DRO method, it provides a more conservative solution. This figure compares system trajectories under the DRO approach with a sample averaging method, showing that DRO yields better adherence to the safety bound. In this figure the system trajectories are shown for different realizations for the nominal system (black), ECP solution using the sample average approximation (green), and the proposed DRO solution (blue). The unsafe boundary is indicated by a red dashed line, and the target area is represented by a red circle.

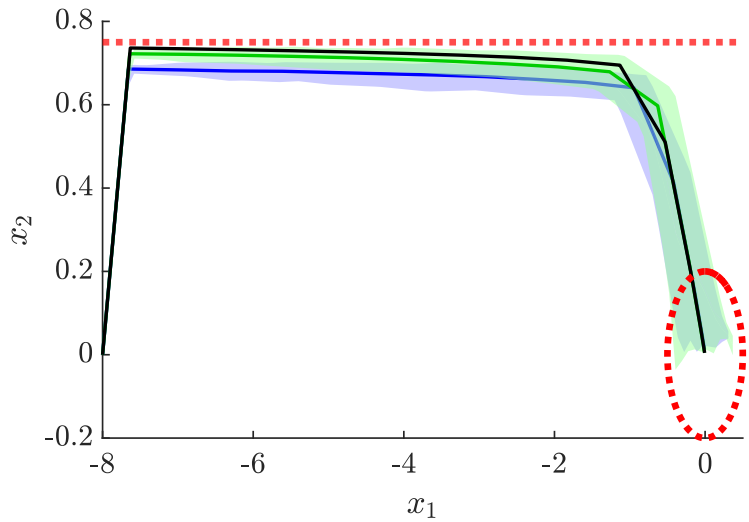

Figure 3.7: System trajectories for different realizations under STL task.

3.4 Conclusion

In this chapter, we explored stochastic linear multi-agent systems under Signal Temporal Logic (STL) specifications and task planning under limited information. By leveraging system linearity, we constructed a probabilistic reachable tube (PRT) at the specification probability level, allowing us to relax the stochastic control problem with a deterministic one with tighter constraints. This PRT-based approach reduces conservatism compared to methods relying on STL structure and is scalable through an algorithm that decomposes the multi-agent problem into iteratively solvable subproblems. However, the conservatism introduced by PRTs grows with the specification horizon, a challenge we aim to address with future data-driven methods.

We also highlighted techniques for handling uncertainty via distributionally robust optimization (DRO) when the underlying probability distributions lie in ambiguity sets. DRO ensures robustness by optimizing over ambiguity sets, such as Wasserstein balls, to handle uncertainty in the probability distribution. Both approaches provide effective strategies for managing limited information, ensuring robust performance and adaptability in uncertain environments

4. Conclusions

This report presented a brief discussion on task planning using limited information, focusing on key techniques and frameworks for enhancing decision-making in multi-agent systems. The exploration was grounded in the need to manage and adapt to incomplete or evolving data, which is crucial in dynamic environments where full information is often unavailable.

We began with an overview of spatiotemporal awareness in task planning, emphasizing the significance of Signal Temporal Logic (STL) for defining and managing complex, time-sensitive tasks. STL's robust semantics provide a structured approach to handle spatial and temporal constraints, which is essential for coordinating multi-agent systems effectively. The report explained how STL enables the encoding of intricate specifications and ensures that agents adhere to these requirements despite challenges posed by limited information.

Following this, we discussed the impact of communication and sensing constraints on multi-agent task planning. Effective coordination in such systems often requires overcoming barriers like restricted communication bandwidth and sensing limitations. Strategies such as decentralized planning and robust task-planning algorithms were identified as crucial for managing these constraints. Decentralized control synthesis was particularly noted for its ability to enhance scalability and flexibility by allowing agents to make decisions based on local information while still contributing to global objectives.

To address control synthesis for stochastic multi-agent systems under stochastic noise and STL constraints, we employed a tube-based constraint-tightening approach allowing the relaxation of the underlying stochastic control problem into a deterministic one with tighter constraints. This method reduces conservatism compared to approaches relying on STL structure. To improve scalability given the complexity of multi-agent systems under STL, we proposed an algorithm that decomposes the problem into iteratively solvable subproblems. The effectiveness of the approach was demonstrated on a ten-agent system.

In addressing optimization under uncertainty, we also explored Distributionally Robust Optimization (DRO). DRO offers a powerful framework for managing stochastic uncertainty by considering ambiguity in the underlying probability distributions. By defining ambiguity sets, such as Wasserstein balls, DRO ensures robustness in decision-making processes. This approach provides probabilistic guarantees and allows for tractable solutions even with limited data, making it a powerful tool for real-world applications where exact distributions are unknown.

As part of our ongoing efforts, we are addressing both theoretical and numerical challenges associated with chance-constrained STL in multi-agent settings, with the aim of publishing our findings in a prestigious journal. Additionally, we are actively working on software development for different control methods and case studies, striving to create a unified tool for the project in relation to WP1. Furthermore, in connection with WP5, we are collaborating with NLR on intent

awareness for unmanned aircraft system traffic management. This collaboration involves employing advanced control methods such as model predictive control and dynamic programming to enhance the effectiveness and safety of traffic management systems.

In summary, the techniques discussed offer efficient solutions for task planning with limited information. These methods ensure that multi-agent systems can maintain high performance and adapt to uncertainties while achieving their goals. Future research should continue to refine these techniques and explore their applications in increasingly complex and dynamic settings.

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